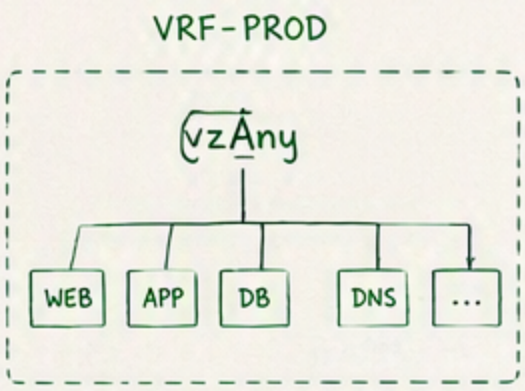
ACI NOTES – vzAny CONTRACTS & SERVICE GRAPH (PBR)

1. What is vzAny ?

- vzAny is a virtual EPG representing ALL EPGs inside a VRF.
- It allows one contract to apply to every EPG in that VRF.
- It does NOT allow traffic by itself. A contract must be attached.



All EPGs in this VRF are members of vzAny

2. Why Use vzAny ?

- Without vzAny – many contracts needed
  - WEB → DNS
  - APP → DNS
  - DB → DNS
  - NTP → DNS
  - ... (50 EPGs = 50 contracts)
- With vzAny – only one contract

vzAny → DNS

(1 contract)

Use Cases:

- ✓ DNS / NTP / AD
- ✓ Shared Services
- ✓ Firewall Insertion
- ✓ Internet Access
- ✓ Transit Routing
- ✓ Migration (Permit All)

Memory Trick:

vzAny = Building Pass (Every EPG inherits it)

3. Contract Relationship (Key Concept)

Consumer EPG (or vzAny) → CONTRACT (Filter + Subject) → Provider EPG (or vzAny)

Result:

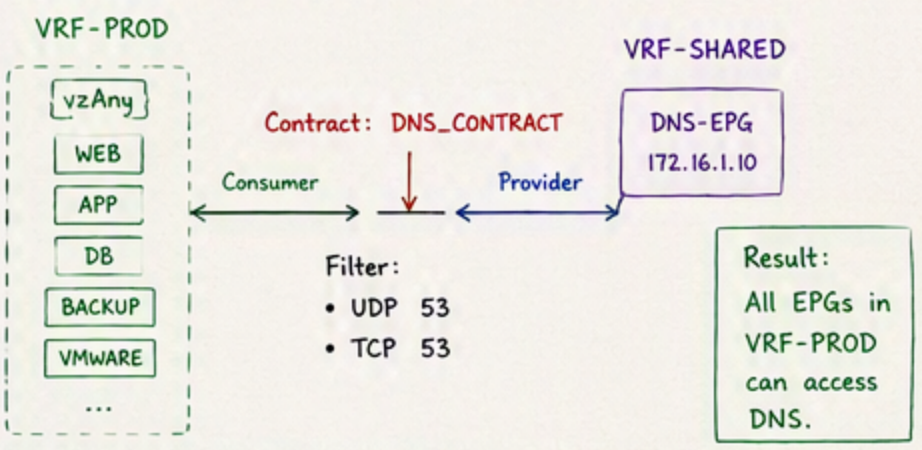
- Consumer ✓
- Provider ✓
- Contract ✓

= Communication ✓

Traffic allowed only when CONSUMER consumes and PROVIDER provides the same contract.

If any one is missing → Traffic Denied

4. vzAny with Example (Shared Services)



Result: All EPGs in VRF-PROD can access DNS.

5. How Traffic is Matched (Internally)

① Packet arrives on Ingress Leaf.
② Source EPG identified.
③ Destination EPG identified.
④ Leaf checks for direct contract.
⑤ If not found → check vzAny membership.
⑥ If matched → contract applied.
⑦ Zoning Rule programmed → Forward.

Example:

Traffic: WEB → DNS

No direct contract WEB->DNS

↓

WEB is member of vzAny

↓

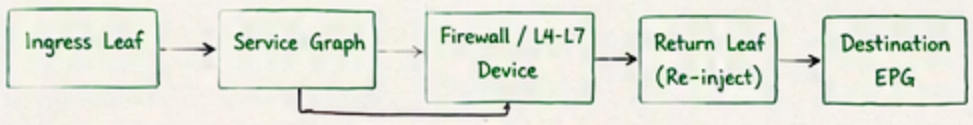
vzAny has contract with DNS

↓

Traffic Permitted

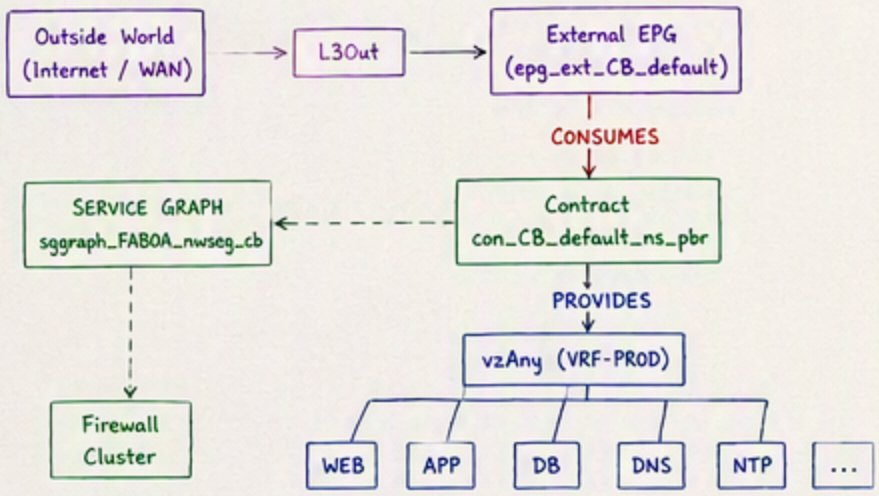
6. Service Graph (PBR) – What & Why ?

- Service Graph redirects traffic to a L4-L7 device (Firewall, LB, IDS, etc) for inspection.
- Implemented using PBR (Policy Based Redirect).
- ★ Contract match → Redirect to Service Graph.

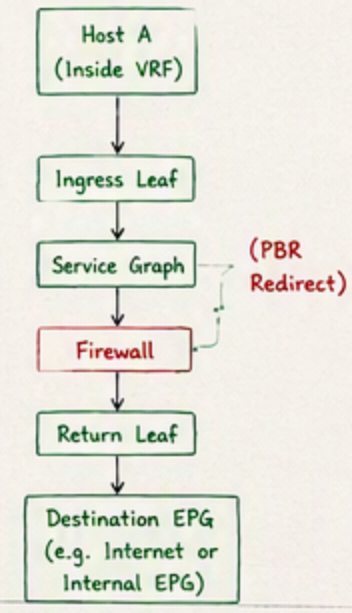


Traffic Flow: Ingress → Service Graph → Device → Return Leaf → Destination

7. Complete Example (Your Environment)



① Host A sends traffic to External IP.
② Leaf receives → Contract check.
③ Contract (con_CB_default_ns_pbr) matched.
④ PBR redirect → Send to Service Graph.
⑤ Traffic forwarded to Firewall.
⑥ Firewall inspects traffic.
⑦ Firewall returns traffic to fabric.
⑧ Return Leaf receives.
⑨ Normal forwarding to destination EPG.



9. Where to Check in APIC GUI

① Tenants → <Tenant> → Networking → VRFs → <VRF>
② Open VRF → click "EPG/ESG Collection for VRF" (This is vzAny)
③ Check Consumed Contracts (what vzAny consumes)
④ Check Provided Contracts (what vzAny provides)
⑤ To see full details → Tenants → Contracts → <Contract>
→ Check Consumers / Providers / Subjects / Service Graph
⑥ Use "Show Usage" to see which nodes & policies use it.

CLI Useful Commands:

- moquery -c vzAny
- moquery -c vzRsAnyToCons
- moquery -c vzRsAnyToProv
- show zoning-rule
- show zoning-rule scope <vrf>

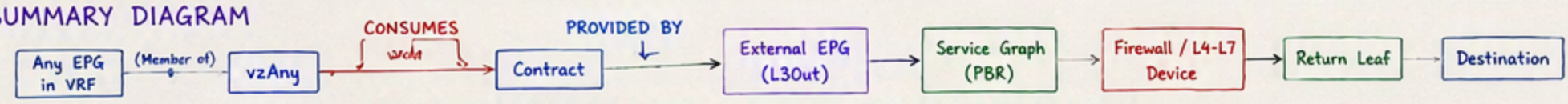
10. Important Points (Golden Rules)

☆ vzAny is NOT a contract. It is a virtual EPG for all EPGs in a VRF.
☆ A contract must be both Consumed and Provided to take effect.
☆ vzAny works only within the same VRF.
☆ Preferred Group is different from vzAny.
☆ Routing + Contract = Communication in ACI.
☆ Without contract, traffic is dropped even if routing exists.
☆ Service Graph is used for PBR / traffic inspection.
☆ Check "Show Usage" if traffic is not working as expected.

Quick Memory:

ARP = MAC Discovery
COOP = Endpoint Location
VXLAN = Transport
Contract = Permission
Service Graph = Redirect

SUMMARY DIAGRAM



One Contract. All EPGs Covered. Centralized Security & Easy to Manage.